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PI Name:	Mueller, Holger Ph.D.		
Project Title:	QWEP Analysis and Optimization		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
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PI Email:	hm@berkeley.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	510-664-4298
Organization Name:	University of California, Berkeley		
PI Address 1:	Physics Department		
PI Address 2:	366 Le Conte Hall # 7300		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-0001	**Congressional District:**	9
Comments:			
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Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
Contact Email:	ulf.e.israelsson@jpl.nasa.gov		
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COI Name (Institution):	Kohel, James (Jet Propulsion Laboratory) Yu, Nan (Jet Propulsion Laboratory)		
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	The equivalence of gravity and accelerated motion is the basis of general relativity's description of gravity as a result of curved space and time. This description, however, is incompatible with the quantum theories of the electroweak and the strong interactions, in which space and time exist independently of the forces. It is therefore not surprising that unified theories of gravity and quantum mechanics are expected to contain tiny violations of the gravity-acceleration equivalence. Such violations would manifest themselves as variations between the acceleration of free fall for different types of matter. These variations are considered as being among the most promising areas to look for signals arising from a unification of general relativity and quantum mechanics. ESA's quantum weak equivalence principle (QWEP) mission seeks such signals with a strongly improved sensitivity of 10-15 or better. By using quantum interference of ultracold atoms, it will use quantum particles where previous tests used bulk matter.		

Task Description:	The research proposed here will combine theory and experiment to maximize the sensitivity and scientific impact of QWEP, and minimize the risk. Using the standard model extension (SME) as a theoretical framework, we will study which species of atoms, what quantum states, and what type of analysis might lead to the largest and most interesting signals. The choice of species, however, is equally dependent on experimental considerations. Using our existing atom interferometers, we will experimentally compare different technologies for the matter-wave beam splitters that are suitable for the candidate species, and develop strategies to minimize systematic effects such as rotation and gravity gradients. Based on the experience of Jet Propulsion Laboratory (JPL) and the Space Sciences Laboratory (SSL) at UC Berkeley, we will match these technologies to the requirement of space operation. Thereby, this work will apply fundamental theory, atomic physics experiment, and principles of astronautics to a comprehensive study optimization and analysis of the QWEP mission.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We seek to understand two things: What is the difference between a quantum and classical test of the weak equivalence principle? What kind of quantum test would provide the most new information regarding Einstein's principle of equivalence? We have made progress on answering these questions have been pursued both theoretically and experimentally.
Task Progress:	The weak equivalence principle (WEP) holds that the acceleration of free fall is the same for all objects, no matter what they are made of. WEP is an integral part of the stronger Einstein equivalence principle (EEP) underlying general relativity, which also requires the universality of the gravitational redshift (UGR) and local Lorentz invariance (LLI). There has been substantial debate in the community regarding whether quantum tests of the weak equivalence principle (QWEP) are more akin to gravitational redshift tests or to classical WEP tests. Central to this controversy is the question of whether a quantum wavepacket of an atom can serve as a "clock" in a gravitational redshift measurement. We have experimentally demonstrated that this is indeed possible, using an atom interferometer to lock the frequency of a quartz oscillator to a subharmonic of the characteristic frequency of a 133Cs atom. This work was published last year in the journal Science [1]. This finding is of particular significance for the planned QWEP experiment on board the International Space Station, as its targeted precision is only modestly better, at 10-14, than classical tests of WEP carried out on Earth, and is less precise than the targeted precision of classical WEP tests in Earth orbit. Though all known models of EEP-violation are such that verifying WEP is equivalent to verifying all aspects of the EEP, and it is conjectured (due to Schiff) that the same is true of any consistent theory, conjecture has never been formally proven. Thus even if WEP is found to be valid to a greater precision, tests like QWEP can help constrain EEP violation in theories that satisfy or violate Schiff's conjecture alike. We have also determined the degree to which existing limits on EEP violation can be improved by a QWEP test, with an eye towards determining which atomic species might be most sensitive to as yet untested forms of EEP violation. The first step was to determine the level at which EEP violation is bounded by existing experiments. Although it is known, e.g., that gravity acts identically upon Titanium and Beryllium in the Earth's gravitational potential at the level of parts in 1013 [2], and upon Cesium and a glass corner cube at the level of parts in 109 [3], corresponding limits on the behavior of elementary were less well understood. Earlier analyses [4-6] of EEP violation using the Standard Model Extension had shown that anomalous gravitational acceleration of conventional matter, i.e. of protons, neutrons, electrons, and their antiparticles, could be no greater than a few parts per million of the overall acceleration. The disparity between these limits and the precision of individual EEP tests [2,3] is due to the fact that most atoms have a fairly similar number of protons, neutrons, and electrons per unit mass, and that all experiments were performed using matter, and not antimatter. Thus much of the signal of EEP violating physics in a given experiment is canceled because anomalies are common to both test masses, and many experiments measure nearly identical combinations of terms governing EEP-violation in elementary particles. This year, we have demonstrated that the total signal for EEP violation in a test body depends not just on what it is made of, but also on how its constituent particles are bound together, and in particular on the bound kinetic energy within a given particle. These corrections are particularly significant for protons and neutrons in atomic nuclei, and enhance the sensitivity of EEP tests using normal matter to anomalies that primarily affect antimatter particles. Using a Woods-Saxon model, we have completed a broad survey of the bound kinetic energy of protons and neutrons in nearly all stable, and many long-lived atoms' nuclei, and applied our findings to determining limits on EEP violation due to existing experiments. We found that three of five observable forms of EEP violation are presently bounded at the level of hundreds of parts per billion, and that one combination of these terms must be less than 21 parts per billion. Our analysis shows that QWEP, which is expected to test EEP for two Rubidium isotopes (85Rb relative to 87Rb) at the level of 10-14, is sensitive to a combination of terms that is relatively orthogonal to those measured by other experiments. Our analysis also shows that a Rubidium vs. Potassium test with the same precision would be roughly 5 times more sensitive. This result will aid the QWEP collaboration in determining which test to perform, given the relative intrinsic sensitivity and systematic backgrounds associated with each possible choice. In addition to the science analysis, we are starting to investigate some of the major systematics in the QWEP measurements and possible mitigations. The dominating systematics is the result of non-ideal overlapping of the two test mass clouds of atoms, leading to limited common mode suppression of magnetic field and gravity field gradients. We seek to mitigate some of these issues by both improving the overlapping of atom clouds and experimental techniques to suppress the related systematics. [1] S.-Y. Lan, P.-C. Kuan, B. Estey, D. English, J. Brown, M. Hohensee, and H. Müller, Science, (2013). [2] S. Schlamminger, K.-Y. Choi, T.A. Wagner, J.H. Gundlach, and E.G. Adelberger, Phys. Rev. Lett. 100, 041101 (2008). [3] H. Müller, A. Peters, and S. Chu, Nature 463, 929 (2010). [4] V.A . Kostelecky and J.D. Tasson, Phys. Rev. D 83, 016013 (2010). [5] M. Hohensee and H. Müller, J. Mod. Opt. 58, 2021 (2011).

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Articles in Peer-reviewed Journals	Lan SY, Kuan PC, Estey B, English D, Brown JM, Hohensee MA, Müller H. "A clock directly linking time to a particle's mass." Science. 2013 Feb 1;339(6119):554-7. http://dx.doi.org/10.1126/science.1230767 ; PubMed PMID: 23306441 , Feb-2013
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Articles in Peer-reviewed Journals	Lan SY, Kuan PC, Estey B, Haslinger P, Müller H. "Influence of the Coriolis force in atom interferometry." Phys Rev Lett. 2012 Mar 2;108(9):090402. PubMed PMID: 22463619 , Mar-2012